

FEDSHIELD-ID

Privacy-First Identity Trust Platform for Banking Networks

SYSTEM ARCHITECTURE SPECIFICATION

A detailed technical specifications document detailing the system architecture layers, API endpoints, database schemas, cryptographic handshakes, and machine learning components of the FedShield-ID platform.

Prepared For: Security & Compliance Audit Team

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Classification: Banking Network Confidential

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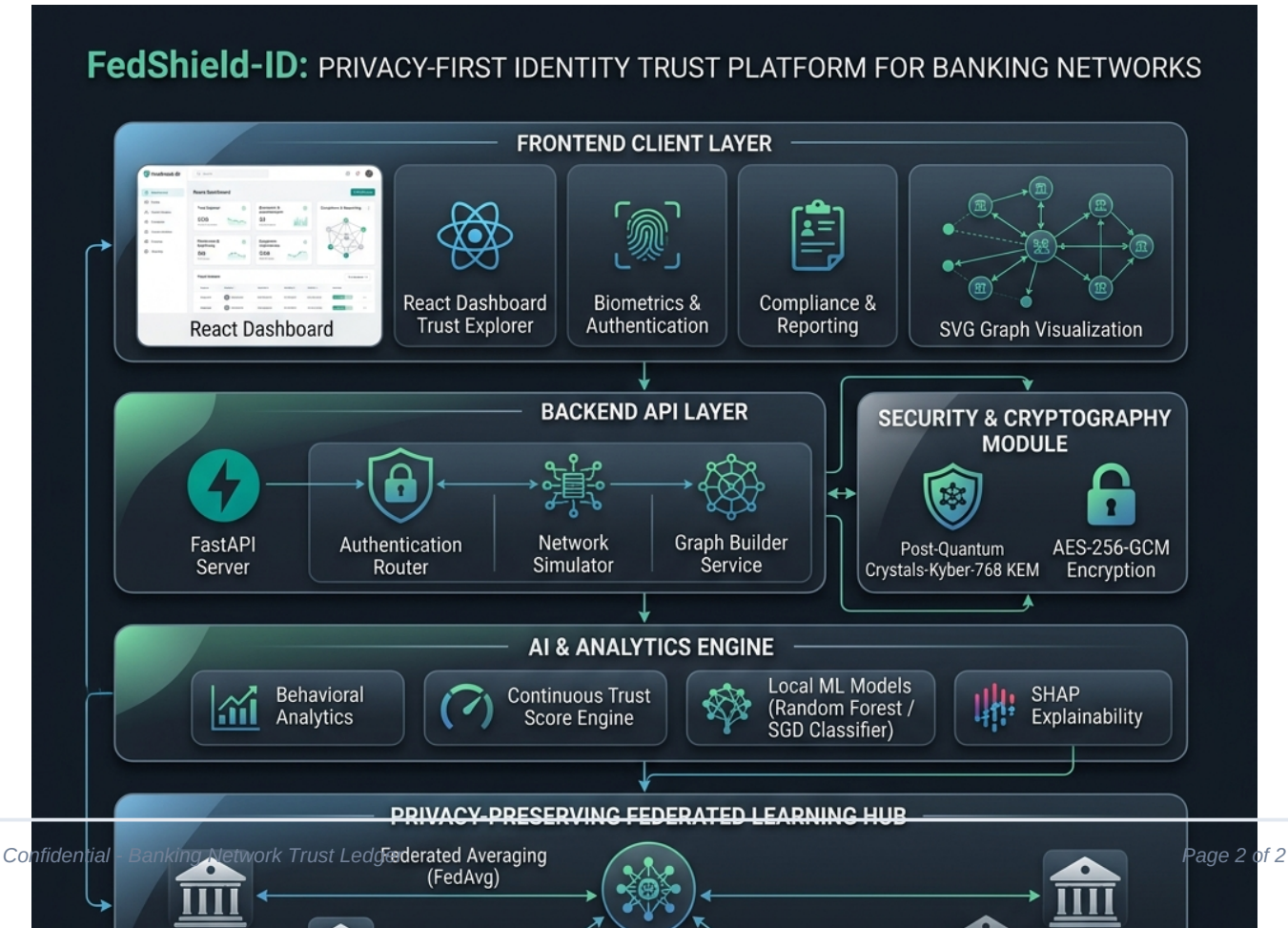
1. Core System Architecture Layers

FedShield-ID implements a highly modular, decoupled architecture consisting of six primary operational layers, designed for sub-millisecond response times during continuous scoring:

- * Frontend Client Layer:** Written in React 18, bundled with Vite. Styled using Tailwind CSS for clean layout. Visualizes biometrics, SHAP attributions, post-quantum benchmarks, and relationship graphs using Chart.js and Lucide React.
- * Backend REST API Layer:** Built on FastAPI, leveraging asynchronous Python handlers. Handles requests, orchestrates background transaction streaming, and coordinates database engines.
- * Cryptographic and Tunnel Layer:** Provides Post-Quantum CRYSTALS-Kyber-768 Key Encapsulation Mechanism (KEM) to securely exchange symmetric keys, protecting model weights during weight aggregation.
- * AI, ML and Explainability Layer:** Maintains local classifiers (Random Forest and Stochastic Gradient Descent), SHAP explainers for transaction scoring, and the GenAI forensic report compiler.
- * Federated Learning Aggregator:** Implements Federated Averaging (FedAvg) across three bank nodes with Laplacian noise injection to enforce Differential Privacy boundaries.
- * Database and Ledger Storage Layer:** Utilizes SQLite for local zero-config developer runs, and PostgreSQL for Docker-based staging environments. Manages logs, profiles, and relationships.

2. System Architecture Diagram

The diagram below represents the modular system architecture and data flows:



3. Database Schema Reference

The system runs on five relational tables and two relationship tables (graph nodes and edges). Below are the schemas for the core tables:

Table: transactions

Field Name	Data Type	Description / Constraints
id	Integer (PK)	Unique transaction identifier
bank	String	Originating node (Bank A, B, or C)
trust_score	Float	Computed Identity Trust Score (0-100)
risk_score	Float	Computed Risk Score (0-100)
device_trust_score	Float	Device health evaluation status
auth_action	String	Adaptive auth verdict (e.g., OTP, BLOCK)
prediction	Integer	Classifier output: 0=Legit, 1=Alert
xai_explanation	Text (JSON)	SHAP values and natural text briefs

Table: user_profiles

Field Name	Data Type	Description / Constraints
customer_id	Integer (PK)	Unique primary customer key
customer_name	String	Customer identifier name
pan_number	String	Permanent Account Number record
avg_typing_speed	Float	Baseline keystroke speed (keys/min)
risk_category	String	Trusted, Low, Medium, High Risk classification

Table: employee_activity_logs

Field Name	Data Type	Description / Constraints
id	Integer (PK)	Auto-increment audit log key
employee_id	String	Identifier of the bank administrator
action	String	Action (e.g., Query PII, Login)
is_suspicious	Boolean	Flags out-of-hours or large database exports

4. Core API Endpoints Reference

FastAPI delivers high-performance routes for frontend queries and transaction streaming. The following represents core routing operations:

SOC and Compliance Feeds

- * **GET /dashboard-metrics:** Aggregates global threat levels, bank data distributions, and overall validation accuracy.
- * **GET /compliance-status:** Audits scores against standard regulations, checking SIM-swaps and insider threat compliance ratios.
- * **GET /privacy-status:** Exposes current epsilon budgets, noise factors, and global model training metrics.
- * **GET /security-status:** Details CRYSTALS-Kyber-768 public keys, payload logs, and execution benchmarks.

Ingestion and Interactive Testing

- * **POST /stream-transactions?active={bool}:** Spawns or halts background threads simulating incoming customer banking events.
- * **POST /simulate-attack:** Triggers deliberate security threats (Insider threat, Credential stuffing, Bot sweep) to verify firewall actions.

Identity Trust and Scoring

- * **POST /predict:** Main route for continuous identity checking. Accepts transaction details, computes behavioral vectors, and feeds database/explainers.
- * **GET /explain/{tx_id}:** Retrieves positive and negative features attribution details computed via SHAP explainer.
- * **GET /fraud-investigation/{tx_id}:** Returns an automated, compliance-ready GenAI investigator text brief.
- * **GET /graph-data:** Retrieves nodes and edges indexing device-sharing and IP clusters to expose collusive fraud rings.